

3. (a) Describe an experiment to investigate the variation of intensity of gamma radiation with the distance from a source. Consider the experimental set-up, data collection and analysis to confirm an inverse square law relationship (space has been provided for a diagram). [6 QER]



- (b) Gamma radiation is known to cause cancer and microwaves are thought by some to be linked to brain cancer.

- (i) Calculate the energy of a photon of:

I. gamma radiation of wavelength 1.0 pm. [2]

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II. microwave radiation of frequency 1.8 GHz. [1]

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- (ii) Some people claim that mobile phone usage increases the rate of brain cancer, while others say that it does not cause significant damage. What steps can be taken to investigate who is correct? [2]

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